Analog electronics

Small signal equivalent circuits of diodes, BJTs and MOSFETs; Simple diode circuits; clipping, clamping and rectifiers; Single-stage BJT and MOSFET amplifiers; biasing, bias stability, mid frequency small signal analysis and frequency response; BJT and MOSFET amplifiers; multi-stage, differential, feedback, power and operational; Simple op-amp circuits; Active filters; Sinusoidal oscillators; criterion for oscillation, single-transistor and op-amp configurations; Function generators, wave-shapingcircuitsand555timers;Voltage reference circuits; Power supplies; ripple removal and regulation.

Digital Circuits

Number systems; Combinatorial circuits; Boolean algebra, minimization of functions using Boolean identities and Karnaugh map, logic gates and their static CMOS implementations, arithmetic circuits, code converters, multiplexes, decoders, encoders, PALs and PLAs; Sequential circuits; latches and flip-flops, counters, shift-registers and finite state machines; Data converters; sample and hold circuits, ADCs and DACs; Semiconductor memories; ROM, SRAM, DRAM; 8-bit microprocessor (8085); architecture, programming, memory and I/O interfacing.

Control Systems

Basic control system components; Feedback principle; Transfer function; Block diagram representation; Signal flow graph; Transient and steady-state analysis of LTI systems; Frequency response; Routh-Hurwitz and Nyquist stability criteria; Bade and root-locus plots; Lag, lead and lag-lead compensation; State variable model and solution of state equation of LTI systems.

Communications

Analog Communications: amplitude modulation and demodulation, angle modulation and demodulation, spectra of AM and FM, super heterodyne receivers, circuits for analog communication; Information theory: entropy, mutual information and channel capacity theorem; Digital communications: PCM, DPCM, digital modulation schemes, amplitude, Gram Schmidt orthogonalization procedure, phase and frequency shift keying (ASK, PSK, FSK), QAM, MAP and ML decoding, matched filter receiver, calculation of bandwidth, SNR and BER for digital modulation; Fundamentals of error detection and correction, Single parity code, Hamming codes; Timing and frequency synchronization, inter-symbol interference and its mitigation; Basics of TDMA, FDMA and CDMA.

Probability and Statistics: Mean, mode, median and standard deviation, distribution function, density function, conditional probability, Baye'stheorem uniform, bionomial, Poisson, normal and exponential distribution, Gaussian distribution function, movements, centralized moments, characteristic functions.

Random Process: autocorrelation and power spectral density, properties of white noise, filtering of random signals through LTI system.

Electro magnetics:

Electro magnetics; Maxwell's equations: differential and integral forms and their interpretation, boundary conditions, wave equation, Poynting vector; Plane waves and properties: reflection and refraction, polarization, phase and group velocity, propagation through various media, skindepth; Transmission lines: equations, characteristic impedance, impedance matching, impedance transformation, S-parameters, Smith chart, Rectangular and circular Waveguide, Waveguide components,klystrons, Magnetron,TWT, microwave solid state microwave devices (Gunn, Tunnel, IMPATT diode), dipole and monopole antennas.

10. Department of Environmental Science and Engineering

Characteristics of water and waste water, Water quality requirements, water &waste water treatment, Air quality, Air Pollution &control, Solid waste, Solid Waste management, Engineering system for resource and energy recovery, Industrial waste management, Environmental impact assessment.

11. Department of Humanities

DISCIPLINE : English

Unit 1: History of English Literature

British Literature from Chaucer to the present day: Age of Chaucer, Age of Renaissance, Restoration Period, Neo-Classical Age, Romantic Age, Modern Age, Post-Modern Age.

Unit 2: History of English Language

Evolution of English Language, Rhetoric and Prosody, English Language Teaching

Unit 3: Literature Theory and Criticism

Plato to T.S. Eliot, Marxist Theory, Feminist Theory, Psychoanalytic Theory, Neo-Historicism, Modernism, Structuralism, Post-Modernism, Post-structuralism, Environmentalism

Unit 4: Classical European and Non-British Literature

European Literature from the classical Age to the 20^{th} century, Indian writing in English and Indian Literature in English translation, American and othernon-British English Literature

The Following Syllabus is common for :

DISCIPLINE : Humanities (Economics)

DISCIPLINE : USME (Economics)

Micro Economics

Consumer behaviour, Demand and Supply analysis, Concept of Elasticity. Theory of production and costs, Forms of Market, Pricing and Output Decision. Tax and Subsidy, Elements of General Equilibrium and Welfare Economics.

Macro Economics

Determination of output and Employment, National Income- Concept and Determinants:

Concept of money, bank, Inflation- Causes and Remedies, Concept of multiplier, Business Cycle, IS and LM function. Concept of Growth and Development: Various models of Growth.

International Trade

Theories of International Trade, Balance of Payment, Terms of Trade, Free trade and Protection.

Indian Economy

Main Features: Geographic Size, Natural Resources, Population, Poverty, Agriculture, Industry, Unemployment, Public finance, Meaning and Measurement of Growth Development-meaning and Characteristics of Underdevelopment.

Statistics

Measures of Central tendency, Measurement of Dispersion, Correlation, Regression, Interpolation and Extrapolation Sampling Distributions Normal, t, Chi square, F distribution, Testing of hypothesis, Index numbers.

12. Mechanical Engineering (25% from each section)

Section 1: Applied Mechanics and Design

Engineering Mechanics: Free-body diagrams and equilibrium; friction and its applications including rolling friction, belt-pulley, brakes, clutches, screw jack, wedge, vehicles, etc.; trusses and frames; virtual work; kinematics and dynamics of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy for mulations; Lagrange's equation.

Mechanics of Materials: Stress and strain, elastic constants, Poisson's ratio; Mohr'scircle for plane stress and plane strain; thin cylinders; shear force and bending moment diagrams; bending and shear stresses; concept of shear centre; deflection of beams; torsion of circular shafts; Euler's theory of columns; energy methods; thermal stresses; strain gauges and rosettes; testing of materials with universal testing machine; testing of hardness and impact strength.

Theory of Machines: Displacement, velocity and acceleration analysis of plane mechanisms; dynamic analysis of linkages; cams; gears and gear trains; fly wheels and governors; balancing of reciprocating and rotating masses; gyroscope.